

NEXL:Elevate Your Learning

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Outline

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- Objective
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- Remaining / Future Tasks



Introduction

NEXL: Elevate Your Learning is a specialized, AI-enhanced Learning Management System tailored for the Nepali educational landscape.

Built on the MERN stack, NEXL bridges the gap between traditional learning and global standards by integrating local payment gateways (eSewa/Khalti) and leveraging advanced AI Transformers for content summarization and personalized recommendations. It empowers independent educators with affordable monetization tools while ensuring students have an interactive, accessible, and high-performance learning environment.



Problem Statement

- Lack of advanced LMS platforms in Nepal
- Infeasible to pay via international online payment provider
- Limited user engagement in existing platforms
- Lack of intelligent features like recommendations and summarization



Objectives

- To develop a full-featured LMS using MERN stack
- To integrate secure local payment systems
- To implement AI-based summarization and recommendation
- To improve user engagement through quizzes, badges, and tracking



Waterfall Model

- **Requirements Analysis**
Defined LMS features and system scope
- **System Design**
Designed architecture, database, and APIs
- **Implementation**
Developed frontend and backend modules
- **Testing**
Verified system functionality
- **Deployment**
Deployed frontend on Vercel and backend services on Render
- **Maintenance**
Updated system features based on user feedback and requirements



Description of Algorithms

- AI Summarization

Uses BART Transformer model to generate concise content summaries

- Recommendation System

Approach content-based filtering

- Payment Integration

E-Sewa API using secure RESTful communication



System Design / Architecture

- Three-tier architecture:
 - Frontend (React)
 - Backend (Node.js / Express)
 - Database (MongoDB)
- RESTful API communication
- Secure authentication and data handling



System Design / Architecture

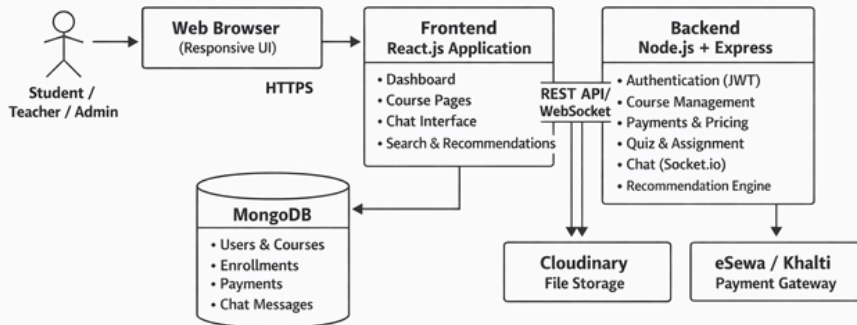


FIG:Block Diagram



Key Features

- Course creation and enrollment
- Secure payment and coupon system
- Quiz system, badges, and progress tracking
- AI-powered summarization and recommendations



Tasks Completed / Progress

- Requirements Phase
- Design Phase
- Implementation Phase:
 - Authentication
 - Course System
 - Payment Integration
 - Recommendation
 - Chat system
- Testing Phase:
 - Chat system
 - Enrollment
 - Payment
- Next Phase:
 - Instructor analytics
 - UI improvements
 - AI Summarization



Implementation & Testing

- Technologies Used:
 - MongoDB , Express , React , Node.js
- Implemented Modules:
 - Authentication
 - Course management
 - Payment system
- Testing:
 - Functional testing of core workflows



Remaining / Future Tasks

- Community / chat system integration
- Mobile responsiveness
- UI/UX improvements
- AI Summarization using BART Transformer model to generate concise content summaries



Thank you

Thank you! Any questions or
feedback are welcome!

